K. K. Wagh Arts Commerce, Science, & Computer Science College

Saraswati Nagar, Nashik-422003

Department of Computer Science

Agriculture Management System

SUBMITED BY:

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K K Wagh Education Society's

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Saraswati Nagar, Nashik-422003

Department of Computer Science

Certificate

This is certified that the student Darshan Shankar Agale Seat No 7057 has successfully completed their project on topic Agriculture Management System as fulfillment of B.Sc. (Computer Science) for the academic year 2024 -2025.

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Project Guide

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HOD

Internal Examiner

External Examiner

ACKNOWLEDGEMENT

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Introduction

Agriculture has been the backbone of human civilization, providing essential resources such as food, raw materials, and economic stability for centuries. However, as populations grow and global trade expands, traditional agricultural management methods struggle to keep up with the increasing demand for efficiency, transparency, and accessibility. Many farmers still rely on outdated, manual processes for managing their crops, fertilizers, and transactions, which often lead to inefficiencies and financial losses.

In today's digital era, integrating technology into agriculture is essential to modernize supply chains, enhance productivity, and ensure fair trade. The **Agriculture Management System (AMS)** aims to address these challenges by providing an **online platform that connects farmers and seller directly, streamlining the sell and purchase of agricultural products.** This system minimizes dependency on intermediaries, enabling **fair pricing, efficient inventory management, and transparent transactions.**

One of the key challenges faced by farmers is **limited access to broader markets**. Due to geographical constraints and lack of information, many farmers are unable to buy their products at competitive prices. The proposed system will act as a **digital marketplace.**

The AMS will incorporate functionalities such as **user authentication, product listing, inventory management, cart handling, order placement, and a secure payment system**. Additionally, an **admin dashboard** will be provided for monitoring transactions, managing users, and ensuring smooth operations.

By implementing this system, we aim to **enhance the agricultural trading experience**, reduce manual efforts, and improve the overall efficiency of the agriculture industry. The system will not only **support farmers and seller** but also contribute to a more **sustainable and profitable agricultural ecosystem**.

1. Problem Definition

Agriculture is a critical sector that sustains economies and livelihoods, yet many of its traditional practices remain inefficient. Farmers and sellers face several challenges that hinder smooth transactions, fair pricing, and effective agricultural trade. The existing manual approach to agricultural transactions leads to **data inaccuracy, time-consuming processes, and market limitations**.

Key Challenges in Traditional Agriculture Management:

1. **Manual Record-Keeping:** Farmers and sellers rely on **paper-based records**, which are prone to errors, data loss, and inefficiency in retrieving information.
2. **Dependency on Intermediaries:** Middlemen often control pricing, leading to **lower profits for seller and higher costs for farmers**
3. **Limited Market Access:** Farmers have difficulty reaching seller **beyond their local markets**, reducing sales opportunities.
4. **No Digital Payment System:** Most transactions are cash-based, causing **delays, fraud risks, and payment disputes**.
5. **No Order Management System:** Farmers and seller lack a structured system to **track orders, deliveries, and payments**.
6. **Lack of Pricing Transparency:** Without market insights, seller are **unable to set competitive prices**, leading to unfair trade.

7. **Physical Marketplace Dependency:** Buyers must **visit markets in person**, making the process time-consuming and inconvenient.

Impact of These Challenges:

- Seller **receive lower profits** due to middlemen controlling prices.
- Buyers **pay higher prices** and have **limited options** due to geographical restrictions.
- The **absence of digital records** makes financial tracking difficult.
- **Delays in transactions** affect the supply chain, leading to **food wastage and financial losses**.
- **High operational costs** due to manual labor and market dependencies.

The Need for a Digital Solution:

To overcome these problems, a **centralized and digitalized** solution is necessary. An **Agriculture Management System** can provide farmers and seller with a **secure, efficient, and transparent platform** to trade directly. By eliminating **manual processes, reducing dependency on intermediaries, and ensuring fair pricing**, the proposed system will **revolutionize agricultural trade and management**.

2. Existing System

The existing agricultural trade system relies heavily on **traditional and manual processes**, which are not well-equipped to handle large-scale trade efficiently. Farmers and seller must depend on **physical marketplaces, intermediaries, and word-of-mouth marketing** to conduct transactions.

How the Current System Works:

1. To sell crops seeds and fertilizers, seller visit **local markets or rely on middlemen** to find buyers.
2. Buyers travel to different markets to find suitable agricultural products.
3. Payments are typically **cash-based**, which increases financial risks.
4. Seller lack a **real-time inventory system**, leading to mismanaged stock levels.
5. Buyers face **limited options and difficulty in comparing prices and quality**.

Challenges in the Current System:

- **High Dependency on Middlemen:**Admin do not get fair prices due to middlemen taking significant commissions.

- **Inconsistent Record-Keeping:** Paper-based records are prone to errors, loss, and inefficiency.
- **Limited Market Reach:** Seller are restricted to selling within **their geographical area**.
- **No Digital Payments:** Seller and farmers must rely on **cash transactions**, which are not always secure.
- **No Online Product Listings:** Buyers cannot search or compare products easily.
- **No Order Tracking:** Once a sale is made, there is no way to track deliveries and payments.

Disadvantages of the Current System:

- **Lack of transparency in transactions.**
- **Difficulty in maintaining accurate financial records.**
- **Time-consuming and inefficient** trade process.
- **Physical dependency on markets limits trade expansion.**
- **High operational costs due to manual labor.**

Due to these limitations, a **modern digital system** is needed to transform agricultural trade, ensuring **better pricing, efficient management, and a streamlined transaction process**.

3. Proposed System

The **Agriculture Management System (AMS)** is a **web-based platform** designed to eliminate the inefficiencies of traditional agricultural trade by **digitizing** transactions and **directly connecting farmers and** seller. The system will provide a **structured, automated, and user-friendly** approach to managing crops, fertilizers, orders, and payments.

Key Features of the Proposed System:

1. **User Authentication:** Farmers and sellers can **register, log in, and manage their accounts securely**.
2. **Product Management:** Admin can **add, edit, and delete** crops and fertilizers.
3. **Search & Filter:** Buyers can **search crops by district, category, and price**.
4. **Cart & Checkout System:** Buyers can **add items to the cart and place orders seamlessly**.
5. **Digital Payments:** Secure **payment gateway integration** for online transactions.
6. **Order Tracking:** Buyers and Seller can **track order status and delivery**.
7. **Admin Dashboard:** Admins can **monitor users, manage inventory, and oversee transactions**.

Advantages of the Proposed System:

- **Removes middlemen**, allowing direct transactions between farmers and seller
- **Provides transparency** in pricing and transactions.
- **Enables online payments**, reducing fraud risks and delays.
- **Increases market reach** by allowing farmers to sell beyond their local area.
- **Enhances inventory management**, reducing stock shortages and waste.
- **Saves time and operational costs** through automation.

4. Requirement Analysis

The **requirement analysis** phase defines the functional and non-functional requirements needed to develop the **Agriculture Management System (AMS)**. This section outlines the **functional aspects, system expectations, and performance considerations** to ensure the system operates efficiently and meets user needs.

4.1 Functional Requirements

Functional requirements define the core features that the system must provide to users, ensuring that it fulfills its intended purpose.

1. User Authentication & Roles:

- Users must be able to **register and log in** securely.
- Different roles: **Admin, Farmer** with specific access rights.
- Passwords should be **encrypted** for security.

2. Crop & Fertilizer Management:

- Seller should be able to **add, edit, delete, and update** crop and fertilizer listings.
- Each crop/fertilizer must include **name, price, quantity, category, and location.**

- Buyers should be able to **view, search, and filter** agricultural products based on different parameters.

3. Search:

- Buyers should be able to **search crops and fertilizers** using filters like **district, category, and price range.**

4. Shopping Cart & Checkout:

- Users should be able to **add crops and fertilizers to the cart.**
- The system should allow users to **modify, remove, and update cart items.**
- Checkout should include **address selection, order summary, and payment options.**

5. Payment Gateway Integration:

- Generate **order confirmation receipts** for both farmers and seller.
- Provide payment status updates (pending, completed, failed).

6. Order Management & Tracking:

- Seller should be notified of new orders.
- Buyers should be able to **track their orders and check delivery status.**

- Admin should have the ability to **update order status (processing, dispatched, delivered).**

7. Admin Dashboard & Management:

- Admins should have access to **view, edit, and delete user accounts.**
- Admin should be able to **monitor transactions and generate reports.**
- Admin should manage **crop categories, pricing policies, and stock availability.**

4.2 Non-Functional Requirements

Non-functional requirements ensure that the system meets quality expectations, including performance, security, and user experience.

1. Security:

- **User authentication** with encrypted passwords (e.g., **hashed using bcrypt**).
- **Secure payment transactions** with SSL encryption.

2. Usability & User Experience:

- The interface should be **simple, intuitive, and responsive**.

- Navigation should be **smooth, with minimal clicks required** for key operations.

3. Performance:

- The system should load **within 3-5 seconds** for optimal user experience.
- Database queries should be **optimized** for faster retrieval and storage of data.
- The system should support **concurrent users (at least 100 active users at a time).**

4. Reliability & Availability:

- The system should have **99.9% uptime** to ensure continuous availability.
- Should automatically recover from **system crashes** or **server failures**.
- Regular **data backups** should be maintained.

5. Scalability:

- The system should be able to **handle increased traffic and data** as the number of users grows.
- The database should support **large amounts of transactions and product listings**.

5. Scope of the System

The **Agriculture Management System** is a **web-based** platform designed to streamline the process of buying and selling crops and fertilizers. It serves as a **bridge between farmers and sellers**, reducing dependency on intermediaries and ensuring **transparent transactions**.

Key Areas Covered by the System

1. Seller's Perspective:

- Seller can **list crops and fertilizers** available for sale.
- Seller can **add, edit, and delete crops and fertilizers** in the system.
- Ability to **update stock and pricing** dynamically.
- **Manage orders efficiently** with order status tracking.
- **Directly connect with buyers**, eliminating middlemen.
- Increased **profits through fair pricing and transparency**.

2. Buyer's Perspective:

- Buyers can **search for crops and fertilizers** based on district and category.
- The system allows buyers to **compare prices and product details**.
- Provides a **cart and checkout process** for easy transactions.
- Buyers receive **order confirmation and status updates**.

System Benefits & Expansion Possibilities

- **Market Expansion**: The system can be expanded to support **more districts and states**.
- **E-commerce Enhancement**: Future updates can include **advanced filtering and better product recommendations**.
- **Data Analytics**: Future integration of **demand analysis and pricing trends** for better decision-making.
- **Sustainability**: Improved market access for farmers, leading to **better agricultural productivity**

6. Feasibility Study

A **feasibility study** helps determine whether the Agriculture Management System is **practical, viable, and beneficial** before full-scale development. It evaluates the system from technical, economic, operational, legal, and scheduling perspectives to ensure its success.

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1. Technical Feasibility

This aspect assesses whether the required technology is available and capable of supporting the system.

✔ **Technology Stack**:

- The system is built using **PHP, MySQL, JavaScript, HTML, and CSS**.
- It runs on **web servers (Apache) and relational databases (MySQL)**.
- Uses **session management** for user authentication.

✔ **Scalability & Performance**:

- The system can **handle multiple users** simultaneously.
- Efficient **database queries ensure fast data retrieval**.
- It can be **hosted on cloud servers** for scalability.

✔ **Security Considerations**:

- Uses **hashed passwords** for authentication.
- Implements **basic payment processing security**.
- Ensures **role-based access control** (Admin, Buyer).

✅**Conclusion**: The technology is widely used, reliable, and suitable for the system's development.

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2. Economic Feasibility

This aspect determines whether the system is **cost-effective** and provides financial benefits.

✔ **Development Cost**:

- Uses **open-source technologies**, reducing software licensing costs.
- **Minimal hardware investment** needed for deployment.
- Can be **self-hosted or hosted on affordable cloud platforms**.

✔ **Cost-Benefit Analysis**:

- **Reduces manual work**, saving operational costs.
- **Eliminates intermediaries**, increasing profits for farmers.

- Provides a **cost-effective** way for buyers to access agricultural products.

✅**Conclusion**: The project is economically viable with long-term benefits for both farmers and seller.

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3. Operational Feasibility

This checks whether users can operate the system effectively.

✔ **User-Friendliness**:

- **Simple and intuitive UI** for easy navigation.
- **Minimal training** required for farmers and seller.

✔ **Efficiency**:

- The system **automates inventory management, order processing, and transactions**.
- Reduces **manual errors in record-keeping**.

✔ **Adoption Readiness**:

- Farmers and seller **can easily transition** from manual methods to the digital platform.
- Admins can **efficiently manage users and inventory**.

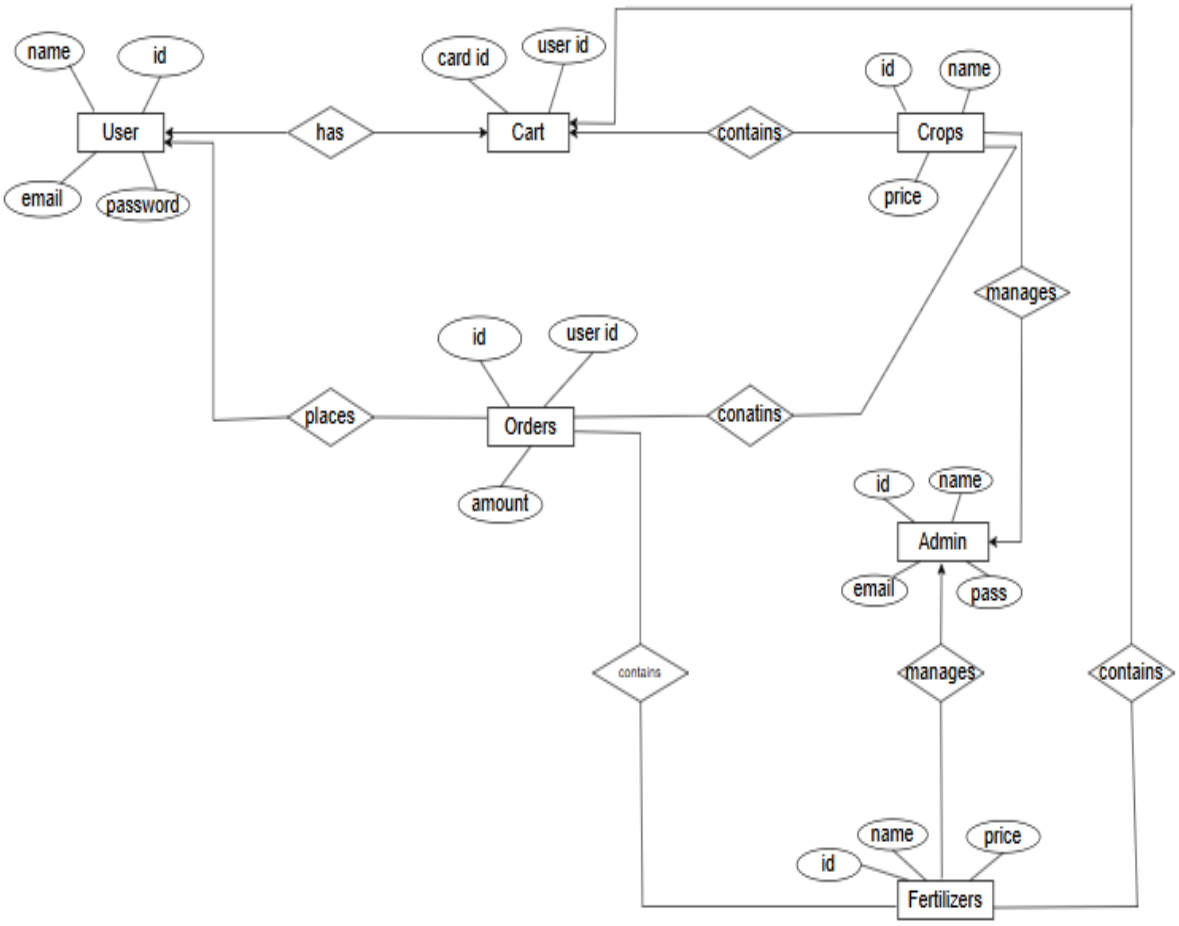
7.Hardware and software Requirements:

Component	Specification
Server	Cloud or Dedicated Hosting with 8GB RAM, 100GB SSD, and a multi-core processor
Client Device	Laptop With Internet access
Network	Broadband/Wi-Fi connectivity (minimum 10 Mbps)

- **Software Requirements:**

Component	Specification
Operating System	**Windows/Linux/macOS**
Programming Language	**PHP, JavaScript, HTML, CSS**
Database	**MySQL**
Server	**Xampp**
Frontend Framework	**Bootstrap/ CSS**

8.ERD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card id
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Cart
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Crops
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Admin
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pass
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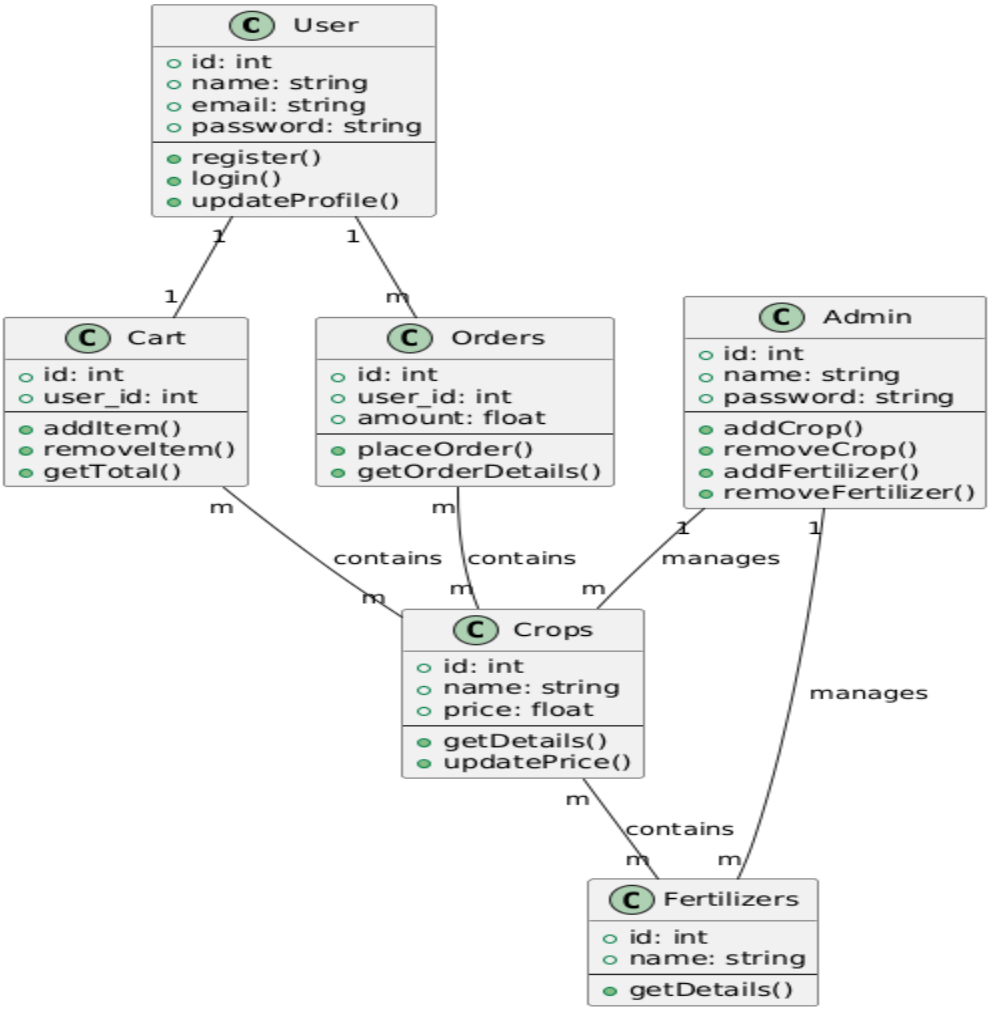
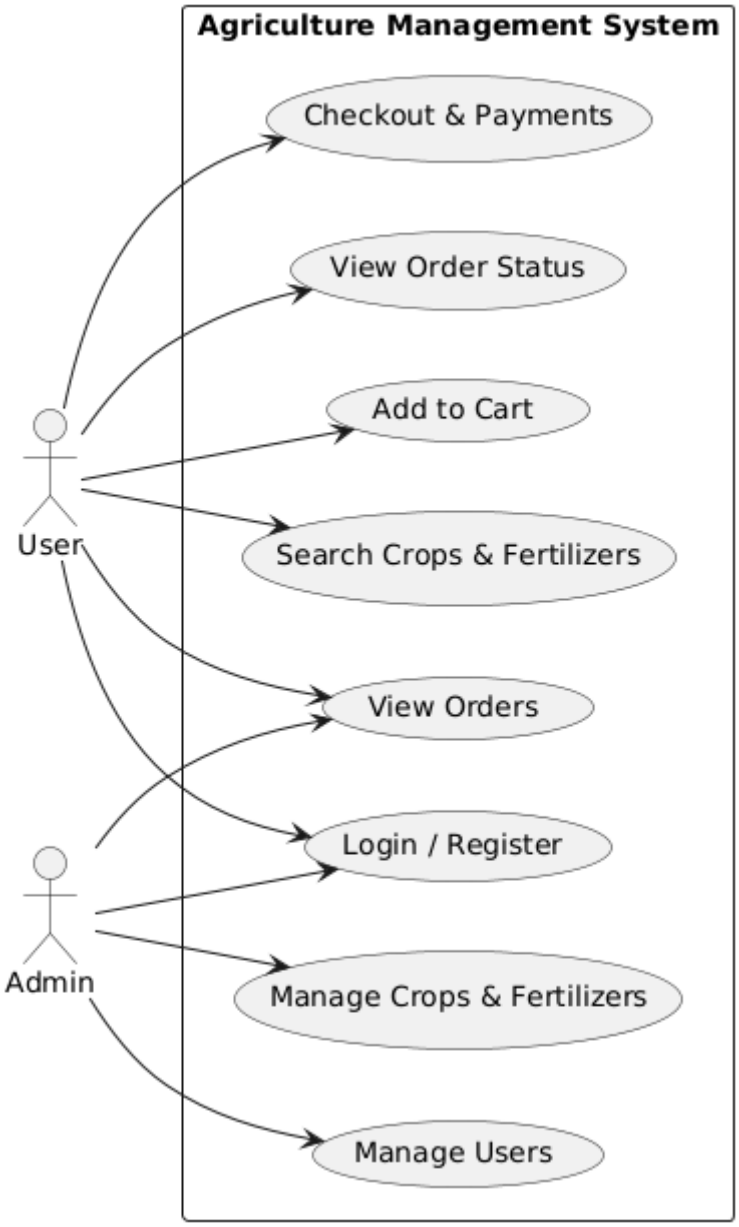
manages
contains
name
price
id
Fertilizers

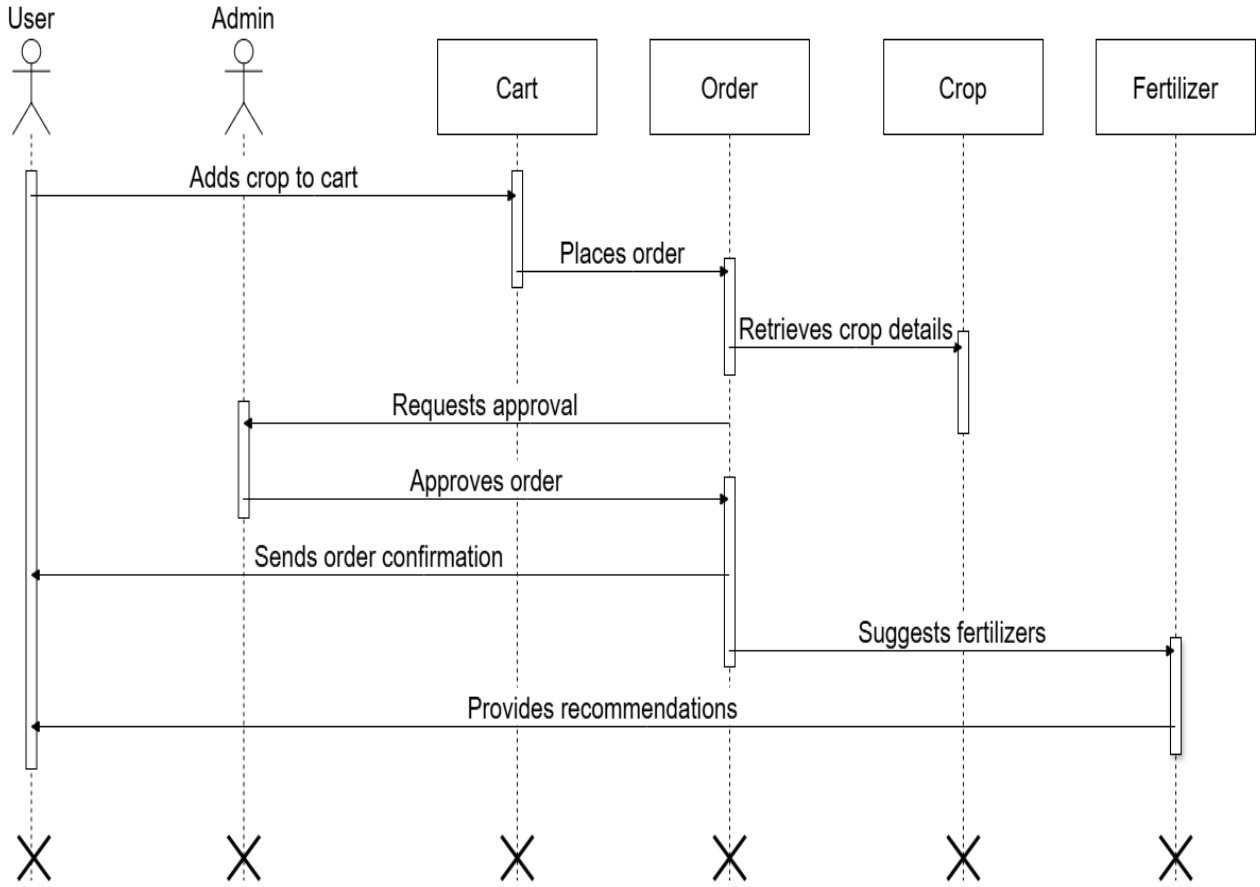
9.UML Diagrams :



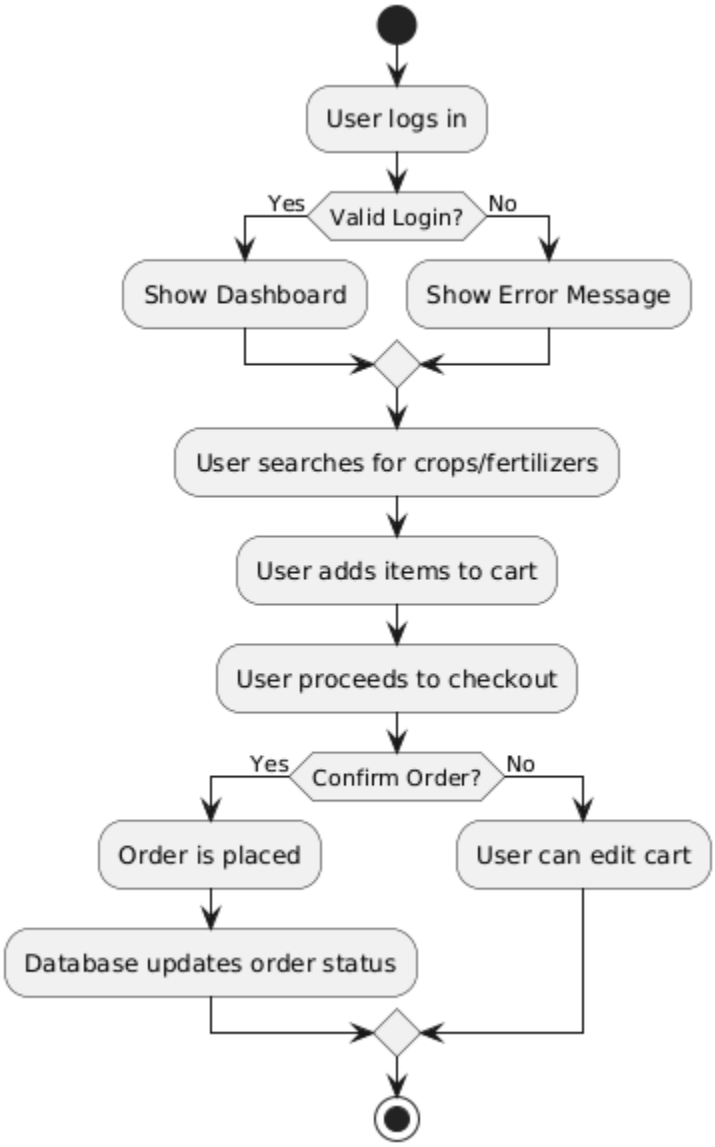
Use case diagram :



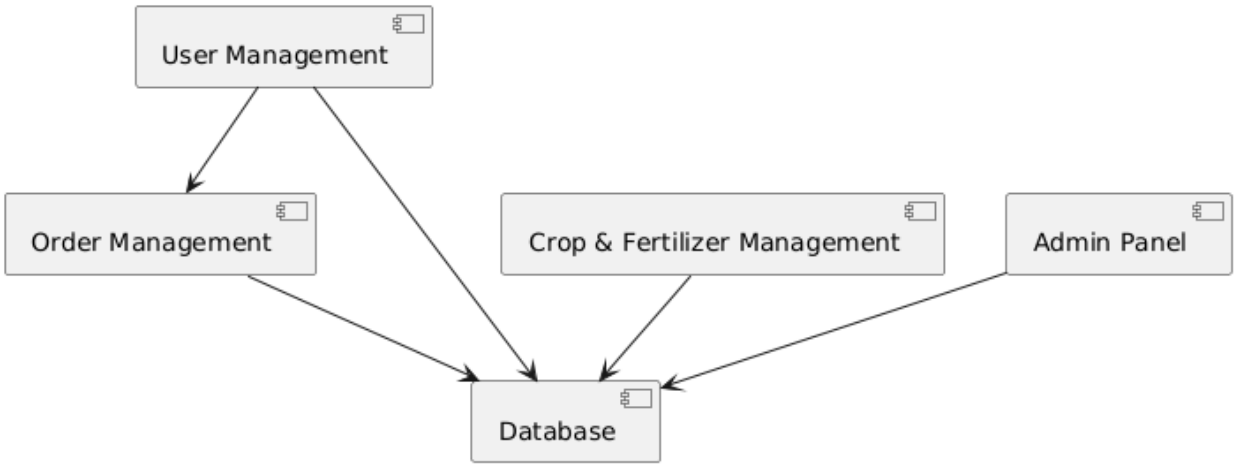
Sequence diagram :



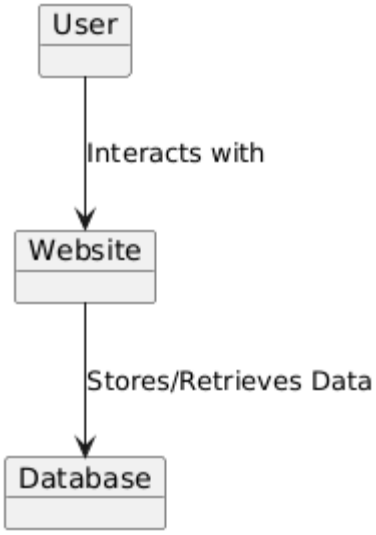
Activity diagram :



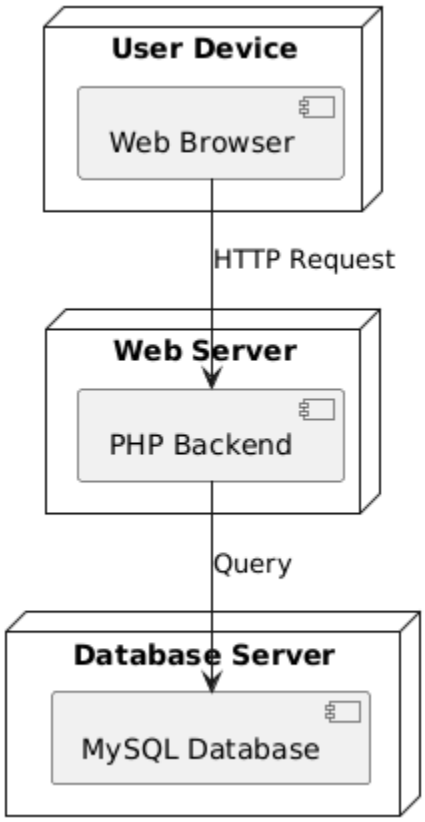
Component Diagram :



Collaboration diagram :



Deployment diagram:



10.Data Dictionary

The **Data Dictionary** defines the structure of the database, including **tables, attributes, data types, constraints, and relationships**. Below is the detailed data dictionary for the **Agriculture Management System**.

Crops table :

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
☐	1	**id**	int(11)			No	*None*		AUTO_INCREMENT	Change	Drop	More
☐	2	**crop_name**	varchar(255)	utf8mb4_general_ci		No	*None*			Change	Drop	More
☐	3	**fertilizer**	varchar(255)	utf8mb4_general_ci		No	*None*			Change	Drop	More
☐	4	**price**	decimal(10,2)			No	*None*			Change	Drop	More
☐	5	**district**	varchar(255)	utf8mb4_general_ci		No	*None*			Change	Drop	More

id-primary key

Fertilizer table:

☐	1	**id**	int(11)		No	*None*	AUTO_INCREMENT	Change	Drop	More
☐	2	**fertilizer_name**	varchar(255)	utf8mb4_general_ci	No	*None*		Change	Drop	More
☐	3	**fertilizer_price**	decimal(10,2)		No	*None*		Change	Drop	More
☐	4	**fertilizer_quantity**	varchar(100)	utf8mb4_general_ci	No	*None*		Change	Drop	More

id-primary key

Orders table:

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
☐	1	**id**	int(11)			No	*None*		AUTO_INCREMENT	Change	Drop	More
☐	2	**user_id**	int(11)			No	*None*			Change	Drop	More
☐	3	**items**	text	utf8mb4_general_ci		No	*None*			Change	Drop	More
☐	4	**total_price**	decimal(10,2)			No	*None*			Change	Drop	More
☐	5	**order_date**	timestamp			No	current_timestamp()			Change	Drop	More
☐	6	**invoice_no**	varchar(255)	utf8mb4_general_ci		No	*None*			Change	Drop	More
☐	7	**payment_mode**	varchar(255)	utf8mb4_general_ci		No	*None*			Change	Drop	More
☐	8	**address**	varchar(255)	utf8mb4_general_ci		No	*None*			Change	Drop	More

id - primary key
user_id - foregin key
invoice_id - unique key

User table:

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
☐	1	**id**	int(11)			No	*None*		AUTO_INCREMENT	Change	Drop	More
☐	2	**name**	text	utf8mb4_general_ci		No	*None*			Change	Drop	More
☐	3	**mobile**	bigint(10)			No	*None*			Change	Drop	More
☐	4	**password**	varchar(8)	utf8mb4_general_ci		No	*None*			Change	Drop	More
☐	5	**address**	varchar(100)	utf8mb4_general_ci		No	*None*			Change	Drop	More
☐	6	**register_as**	varchar(11)	utf8mb4_general_ci		No	*None*			Change	Drop	More
☐	7	**location**	int(11)			No	*None*			Change	Drop	More

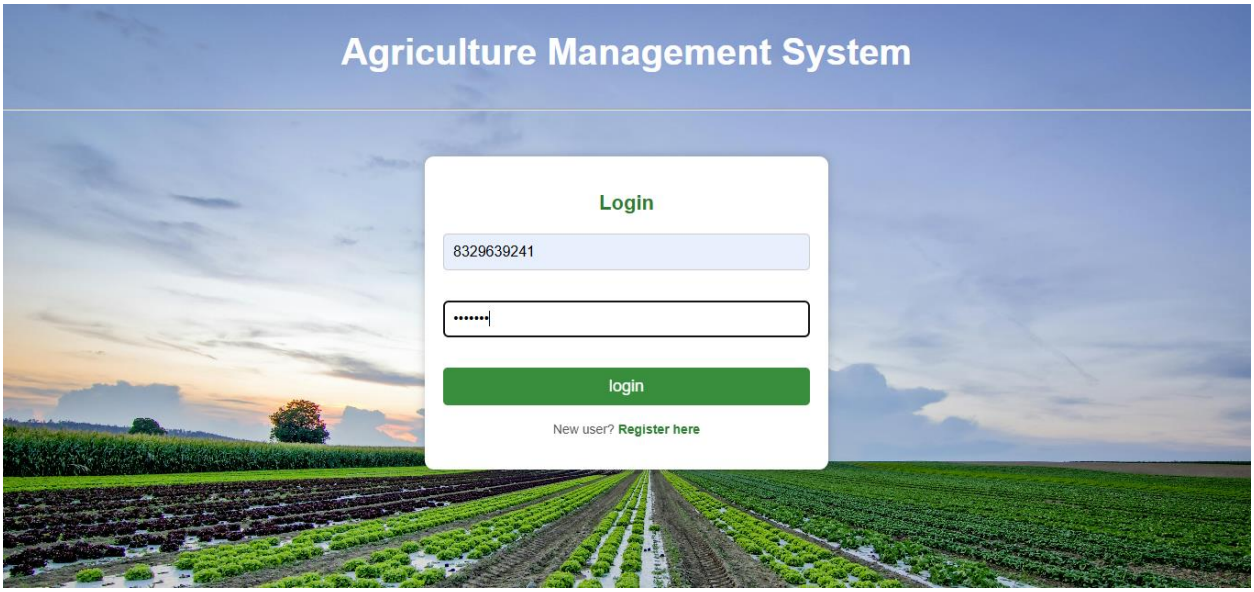
id - primary key

11.Sample I/O screen

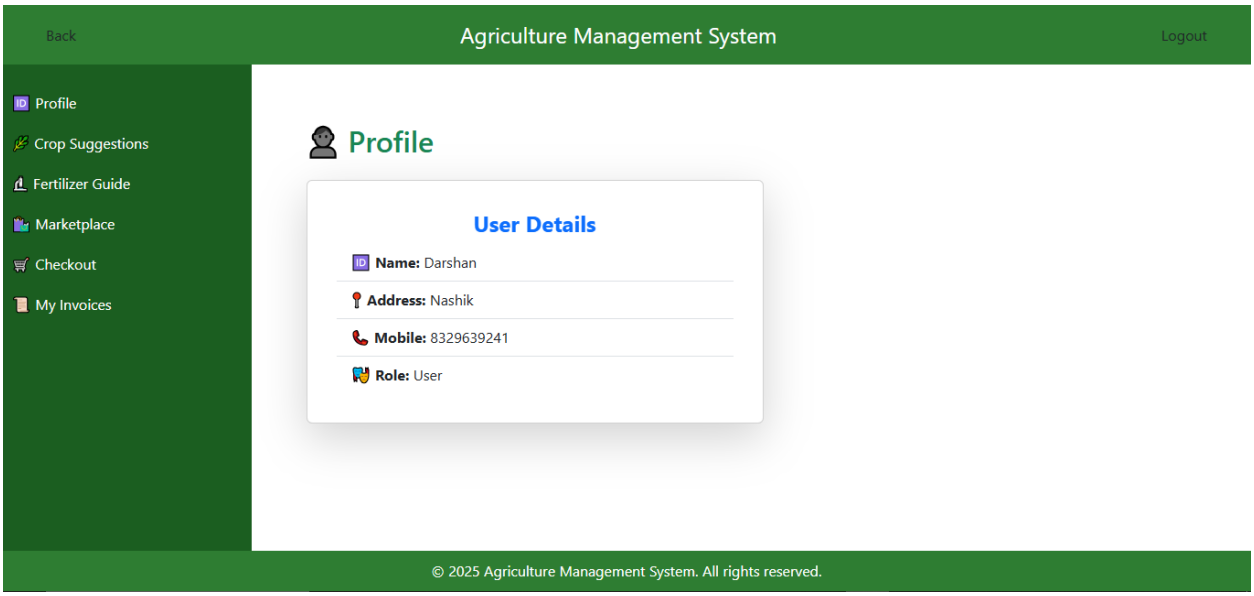
1 . user side

Registration :

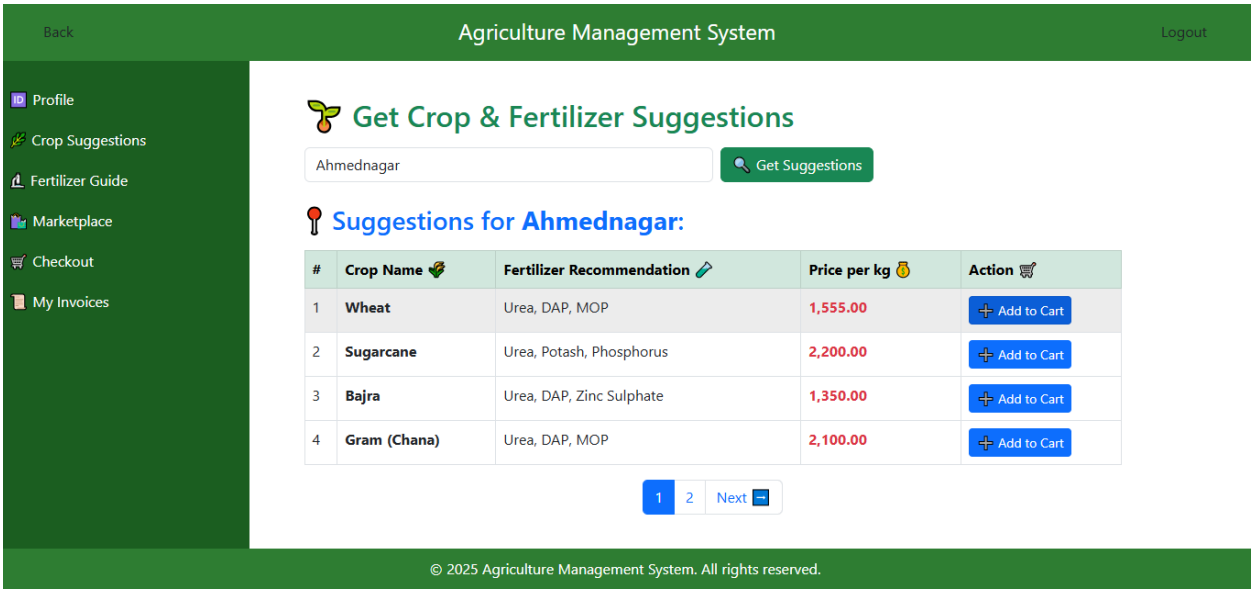
Login:



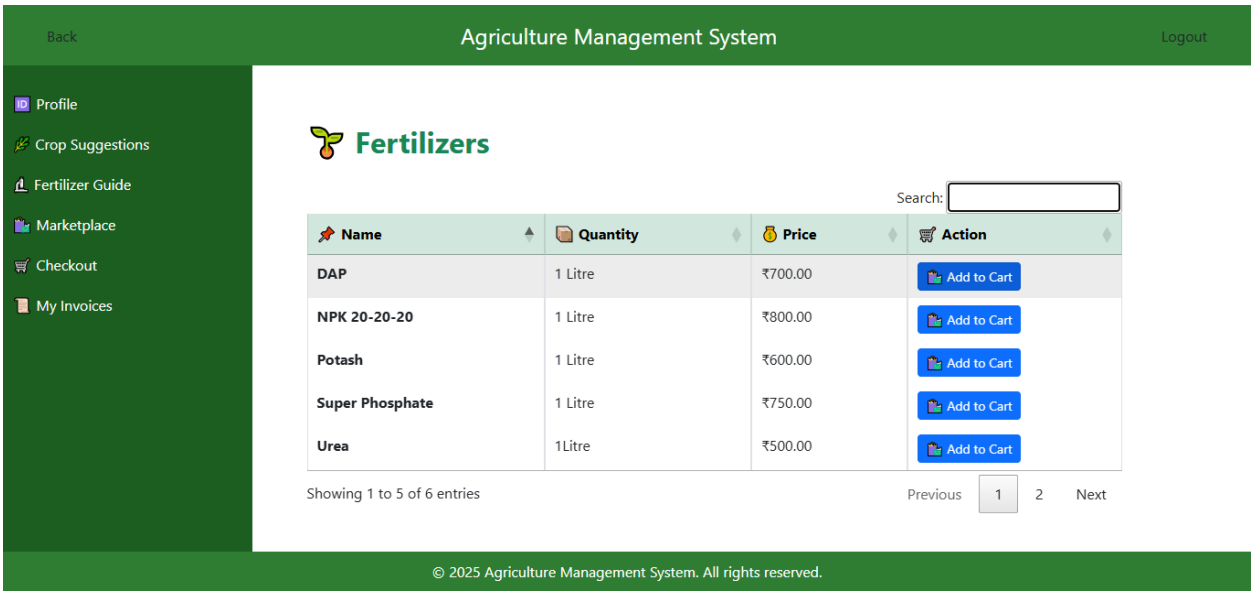
User profile:



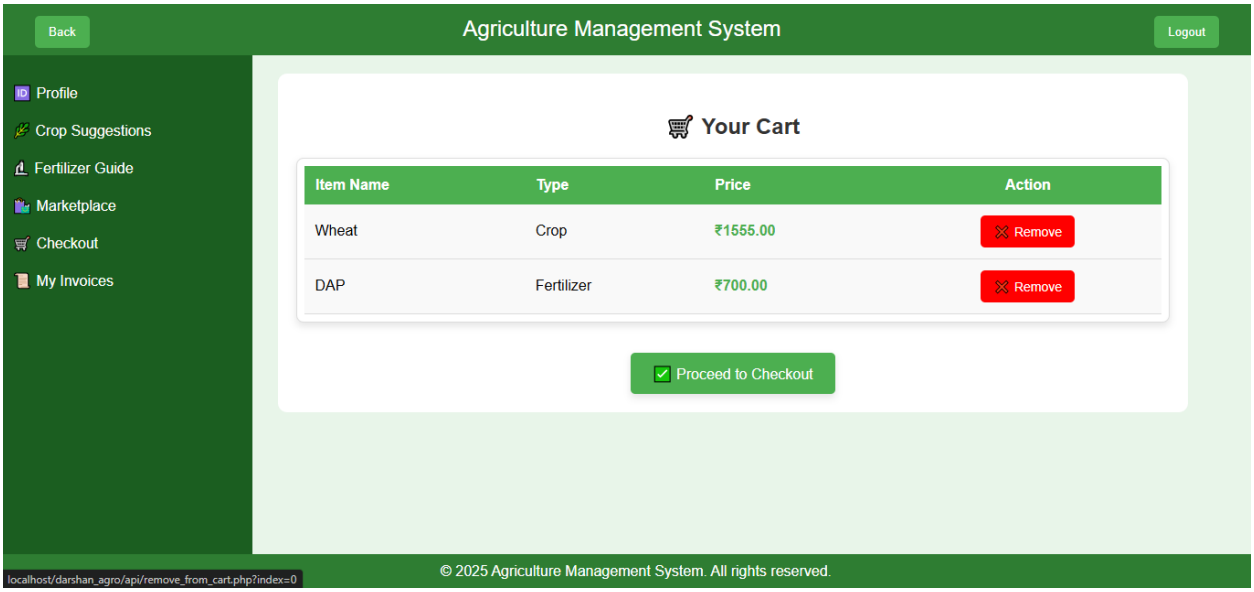
Crops Suggestions:



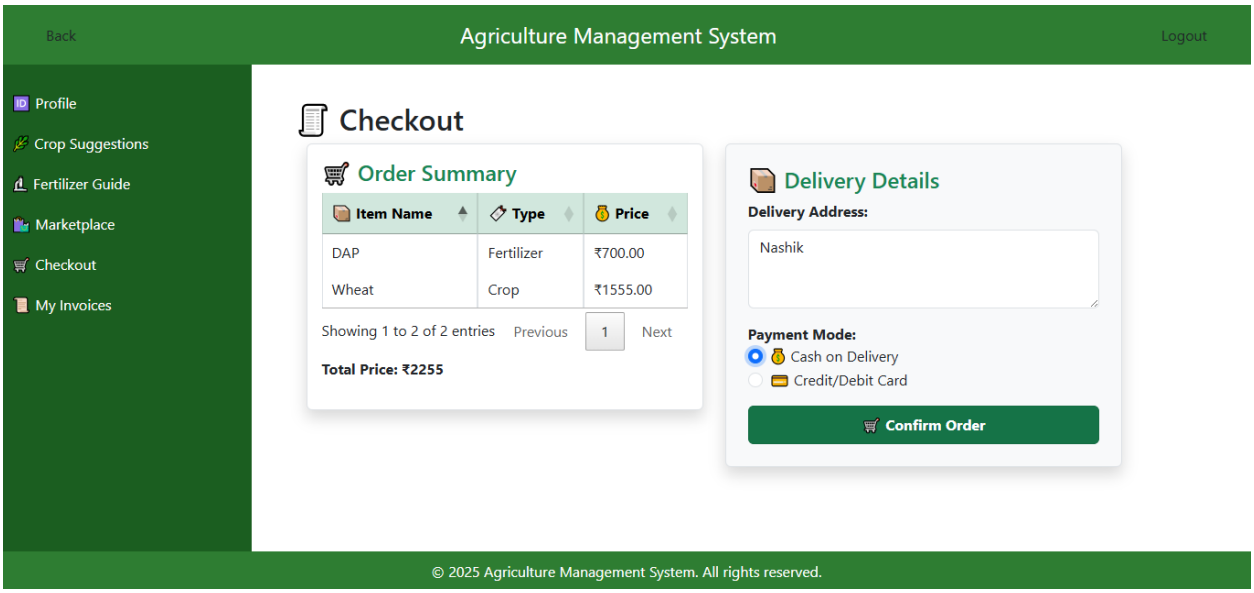
Fertilizer Guide :



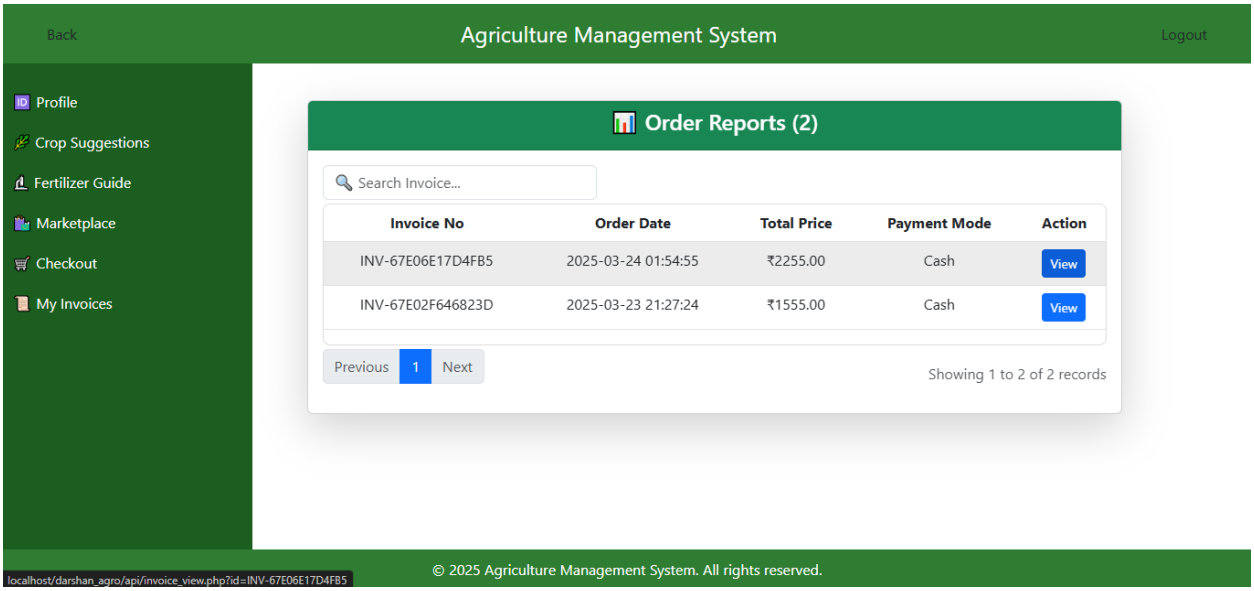
Marketplace:



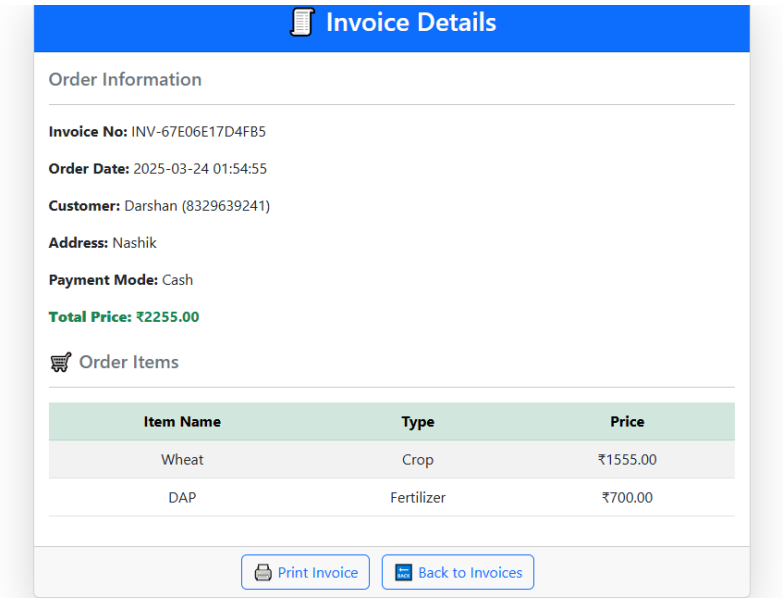
Checkout page:



My Invoices:

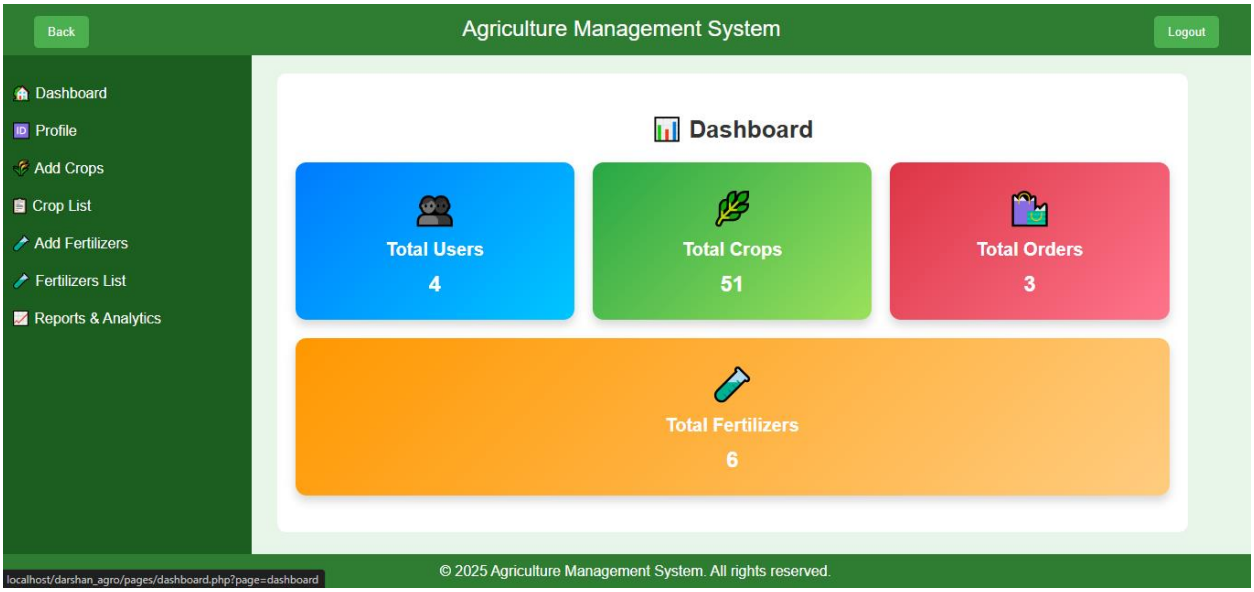


Invoice Details:

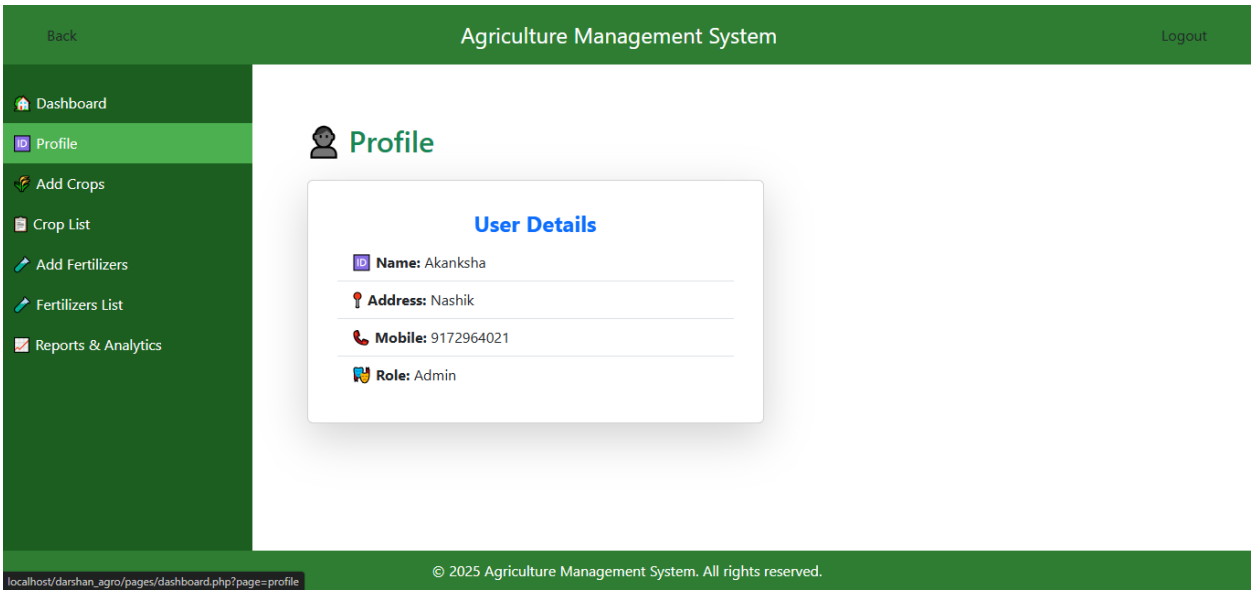


2. Admin side

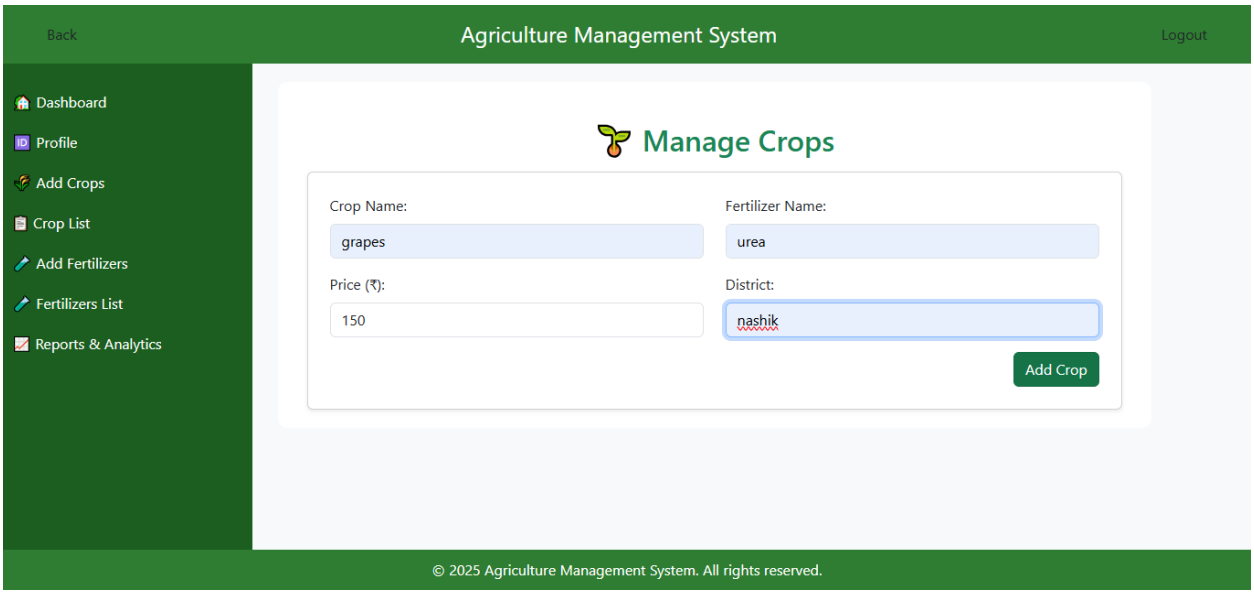
Admin dashboard:



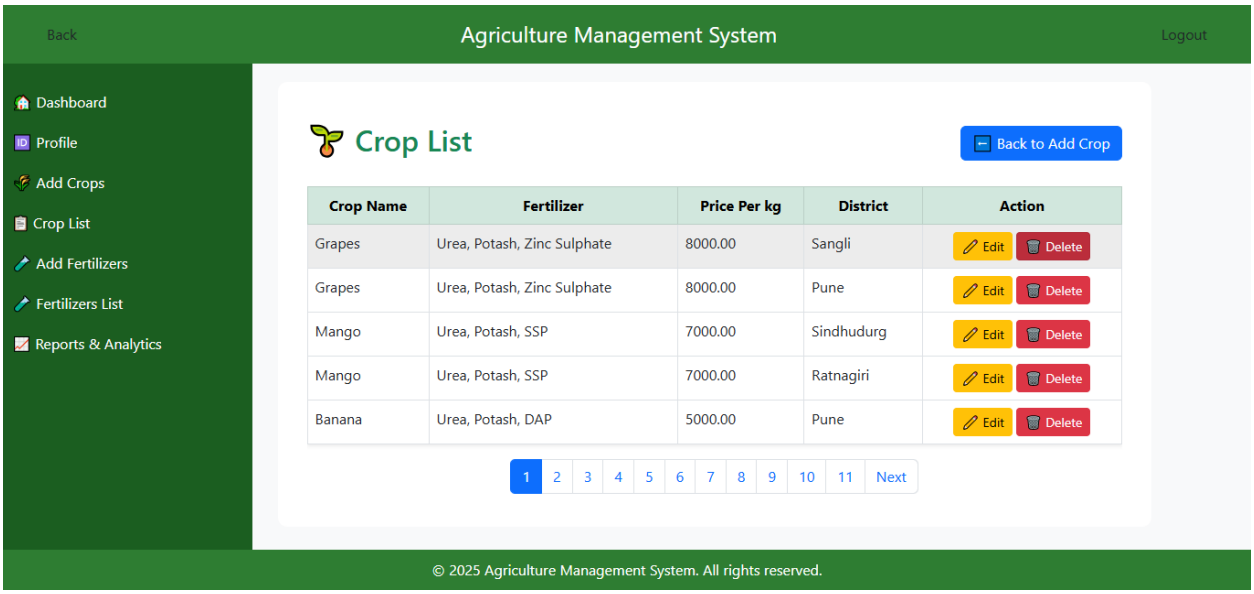
Admin Profile:



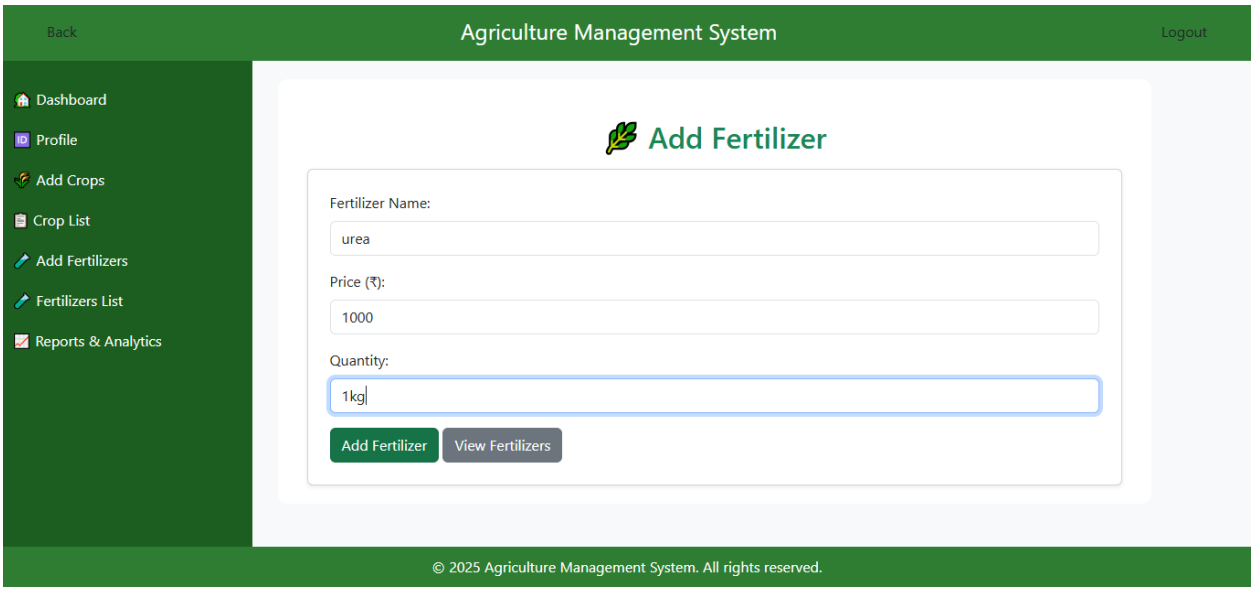
Add Crops:



Crops List:

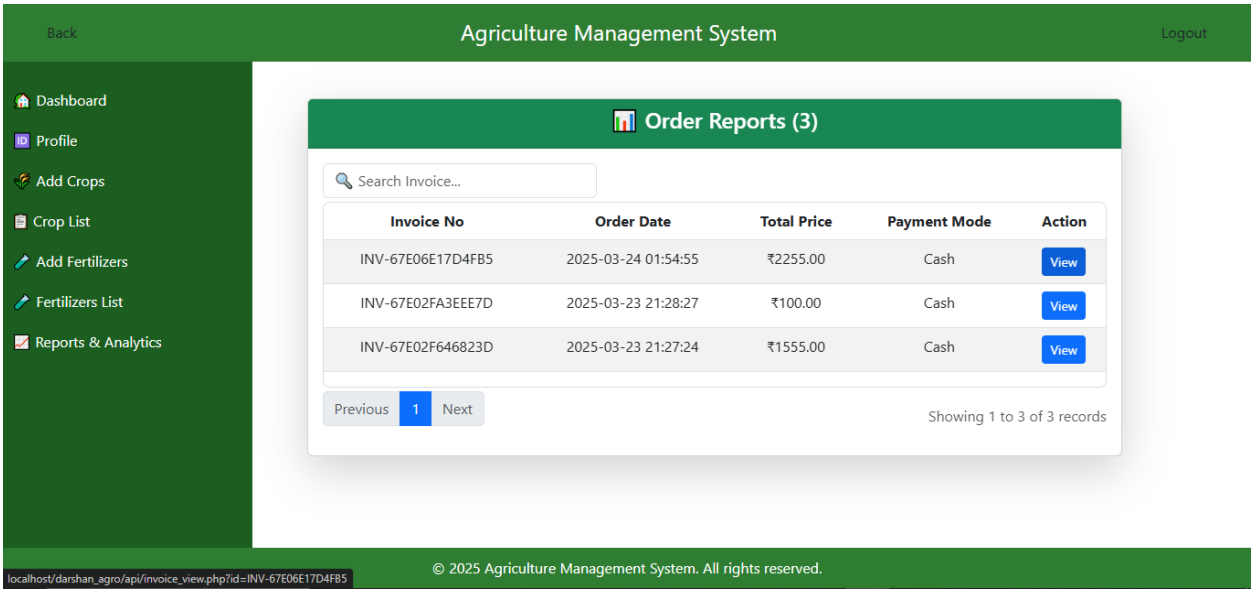


Add fertilizers:



Fertilizers List:

Reports & Analytics:



12.Report Testing

Software testing is crucial to ensure that the **Agriculture Management System** functions as intended, is free of errors, and meets user requirements. This section outlines various testing methodologies applied to different components of the system.

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1. Testing Objectives

The main objectives of testing the Agriculture Management System are:

- To identify and fix errors before deployment.
- To ensure data accuracy and proper functionality.
- To verify that different modules work seamlessly.
- To check system security and performance under various conditions.

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2. Types of Testing Performed

2.1 Unit Testing

Objective: To test individual components of the system, such as user authentication, adding crops, and processing orders.
Approach: Each module is tested separately before integrating it with

other modules.

Example:

- Verify that a user can successfully register and log in.
- Ensure a admin can add, edit, and delete crops and fertilizers.
- Check that an order is placed correctly with the correct price calculations.

2.2 Integration Testing

Objective: To test interactions between different modules.

Approach: The system is tested as a whole to identify any issues in data flow between different parts.

Example:

- Verify that crops added by admin are displayed correctly in the marketplace for buyers.
- Ensure that cart functionality correctly updates the total price when items are added or removed.
- Check that after checkout, the order details are stored correctly in the database.

2.3 Functional Testing

Objective: To test whether the system meets functional requirements.

Approach: Test cases are written based on the functional requirements, and expected results are compared with actual outcomes.

Example:

- Verify that users can search and filter crops based on location.
- Ensure that only admin can add crops or fertilizers.

2.4 Performance Testing

Objective: To evaluate system performance under different conditions.

Approach: Load testing and stress testing are performed to check system stability.

Example:

- Measure system response time when multiple users place orders simultaneously.
- Check database performance with large volumes of data (e.g., 10,000+ crops and orders).

2.5 Security Testing Objective: To ensure the system is protected from security threats such as SQL injection, cross-site scripting (XSS), and unauthorized access.

Approach: Penetration testing and vulnerability assessments are conducted.

Example:

- Attempt to log in using SQL injection to test database security.
- Ensure that users cannot access admin features without proper authorization.
- Test data encryption for storing passwords securely.

2.6 Usability Testing

Objective: To verify that the system is user-friendly and meets user expectations.

Approach: Real users interact with the system and provide feedback on navigation, design, and overall experience.

Example:

- Sellers test the process of adding crops and fertilizers.
- Buyers test searching and purchasing products.
- Admins test managing user accounts, orders, and product listings.

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3. Sample Test Cases : Below are sample test cases used during testing:

Test ID	Test Scenario	Input Data	Expected Result	Actual Result	Status
TC_01	User Registration	Valid name, email, password	User should be registered successfully	As expected	✅ Passed
TC_02	User Login	Correct credentials	User should log in successfully	As expected	✅ Passed
TC_03	Add Crop	Name, price, quantity	Crop should be added	As expected	✅ Passed
TC_04	Search Crops	District name	Relevant crops should appear	As expected	✅ Passed
TC_05	Place Order	Cart items, user details	Order should be placed successfully	As expected	✅ Passed
TC_06	Checkout Process	Crop in cart	Payment should be processed	As expected	✅ Passed
TC_07	Wrong Credentials	Incorrect email/password	Error message should be displayed	As expected	✅ Passed

4. Bug Tracking & Fixing

During testing, several issues were identified and resolved:

- **Issue 1:** Incorrect total price calculation in the cart.

  Solution: Fixed the logic to multiply the unit price by quantity.
- **Issue 2:** Unauthenticated users could access restricted pages.

  Solution: Implemented session-based access control.
- **Issue 3:** Slow loading time for large datasets.

  Solution: Optimized database queries and added indexing.

13. Conclusion

The **Agriculture Management System** successfully digitizes agricultural transactions, making it easier for farmers and sellers to interact without intermediaries. By offering an online platform for crop and fertilizer management, order placement, and transaction tracking, the system enhances efficiency, transparency, and accessibility. The project provides a **centralized and automated** solution to traditional agricultural challenges, improving productivity and market reach for sellers while offering buyers a seamless shopping experience.

The system ensures:

✔ **Reduced dependency on middlemen**, leading to fair pricing.

✔ **Accurate inventory tracking**, preventing stock mismanagement.

✔ **Improved market accessibility**, benefiting both seller and consumers.

✔ **Faster and secure transactions**, ensuring reliability.

By implementing this **user-friendly and scalable** system, the agricultural industry moves closer to technological advancements, streamlining business operations and increasing overall profitability.

✅**Conclusion**: The system is **easy to use, practical, and improves efficiency**.

Conclusion:

By implementing the **Agriculture Management System**, the agricultural sector will experience a **significant transformation** in trade efficiency, profitability, and ease of access. The system will **empower sellers, provide better options for buyers, and ensure a fair and transparent market**

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14. Future Enhancements

While the current system effectively serves its purpose, **several enhancements** can further improve its usability and functionality:

✅**Mobile Application**

- Develop an Android/iOS app to allow sellers and farmers to access the system on the go.
- Enable push notifications for order status updates.

✅**AI-based Crop Recommendations**

- Integrate artificial intelligence (AI) to suggest crops based on **soil type, climate, and market demand**.

✅**Blockchain for Secure Transactions**

- Use blockchain technology for **enhanced transparency and fraud prevention** in transactions.

✅**Advanced Analytics & Reports**

- Generate reports on **sales trends, crop demand, and pricing analysis** to assist sellers in decision-making.

✅**Multi-language Support**

- Provide system accessibility in **regional languages** to accommodate buyers from different backgrounds.

✅**Logistics & Delivery Integration**

- Introduce a logistics module to **track deliveries** and estimate shipping times.

By implementing these **future improvements**, the system can **further revolutionize** the agriculture industry, making it more **efficient, transparent, and buyer-friendly**.

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15. Bibliography

The following resources were referenced for the development of the Agriculture Management System :

 Web Development & Database

- PHP Official Documentation
- MySQL Database Management
- JavaScript & Frontend Technologies

 Agriculture & Market Research

- FAO - Agriculture Market Trends
- Agricultural Marketing & Pricing Research

 Software Engineering & System Design

- Software Development Life Cycle
- Use Case & UML Diagrams

 Security & Payment Process

- Cybersecurity Best Practices
- Online Payment Gateway Systems

These references helped in designing a **secure, efficient, and user-friendly** system.

- "Introduction to PHP and MySQL" – https://www.php.net/
- "Web Development Best Practices" – https://developer.mozilla.org/en/
- "Agriculture Market Trends" – https://www.fao.org/

"E-commerce in Agriculture" – https://www.sciencedirect.com/